

Fly Express Passenger App

System Modules, Feature Catalog & Revenue Architecture Specification

Corridor: **Entebbe ↔ Kampala**
(Express)

Technology: **Vanilla HTML5 / CSS3 / JS**
SPA

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Executive Summary: Fly Express is an express passenger transit, express cargo logistics, private vehicle charter, and digital payment platform built for regional transport corridors. All desktop popups, dialogs, modals, and sheets are strictly constrained to a maximum width of 500px on desktop screens.

1. System Architecture Overview

Overview

The application operates as a high-performance single-page web application (SPA). Core view state management and business logic reside in `script.js`, responsive layouts in `styles.css`, and the semantic container shell in `index.html`. Real-time corridor map routing is provided via Leaflet.js / OpenStreetMap, with canvas dynamic QR code and barcode rendering engines for passenger passes and logistics receipts.

2. Complete Modules Breakdown (14 Modules)

Functional Catalog

1. Authentication, Onboarding & User Setup

USER LIFECYCLE

Manages user entry, account verification, guest sessions, and system-wide accessibility settings.

- Animated splash screen and 3-stage interactive onboarding carousel.
- Passenger sign-in layer with 6-digit OTP verification (123456), phone authentication, and Guest Mode.
- Multi-language translation engine: **English, Luganda, Swahili, and Runyankole**.
- Accessibility manager: High-contrast mode, font scale adjustments, and reduced motion settings.

showOnboarding()

renderOnboarding()

showAuth()

renderAuth()

submitOtp()

updateLanguage()

2. Home Dashboard & Quick Actions

CORE SURFACE

Primary navigation hub presenting live status, countdown timers, and service launchers.

- Time-sensitive greeting system (*Good morning / afternoon / evening*).
- Quick route search shortcut chips (e.g. Entebbe Bus Park → Kampala Railway Stage).
- Live departure carousel with countdown timers, available seat counts, vehicle registration plates, driver photo, and price badges.
- Quick launcher buttons for key services: Book Van, Send Parcel, Special Hire, Wallet Top-up.

`renderHome()``initHomeCarousel()``getAvailableTrips()``getGreetingText()`

3. Passenger Search & Departure Selection

TRIP SEARCH

Enables passengers to search, filter, and compare upcoming vehicle departures.

- Search parameters: Origin stage, Destination stage, Travel date picker modal, and passenger breakdown (Adults, Children, Infants).
- Filterable listings by time, vehicle class (14-seater Commuter, 18-seater Highroof, 30-seater Executive Coaster), price, and seat status (Available vs. Almost full).

`renderBook()``renderSearchResultsScreen()``renderAvailableVansScreen()`

4. Progressive Booking Engine

BOOKING FLOW

A progressive disclosure workflow allowing tailored travel customization.

- **Return Trip Accordion:** One-click toggle adding return journey with discounted rate calculations.
- **Passenger Accordion:** Full name, phone, ID type, and saved passenger fast-selection roster.
- **Assistance & Language Accordion:** Wheelchair access, elderly support, child seats, spoken language preference.
- **Luggage Accordion:** Itemized luggage categories (Personal, Standard, Large, Excess, Fragile, Commercial).
- **Seat Selection Accordion:** "Best Available" mode vs. **interactive 14-seater interior seat map** over vehicle photography (disabled driver seat).
- Interactive OpenStreetMap corridor preview condensing into a sticky map upon scrolling.

`renderTripDetails()``renderPassengers()``renderReturns()``renderLuggage()``initTripMap()`

5. Special Hire (Private Vehicle Charter)

GROUP TRAVEL

Full private vehicle charter booking engine for group travel and custom itineraries.

- 4-step charter wizard: Vehicle Selection → Itinerary & Duration → Driver/Fuel Options → Payment & QR Receipt.
- Fleet options: Saloon Sedan (4 seats), Toyota Noah (7 seats), Minivan (10 seats), Commuter (14 seats), Highroof (18 seats), Executive Coaster (30 seats).
- Custom rates, driver daily allowance, tour guide add-on, fuel fees, and auto-calculated 18% VAT.
- Digital charter pass with scannable booking QR Code (#SH-98402-UG).

renderSpecialHire()

calculateSpecialHirePrice()

renderSpecialHirePaymentPanel()

6. Checkout, Pricing & Multi-Channel Payments

FINANCIAL GATEWAY

Unified checkout and payment authorization engine across all services (modals constrained to ≤ 500 px on desktop).

- Collapsible Shopify-style order summary accordion on mobile; sticky summary sidebar on desktop.
- Itemized cost breakdown (Base fare, seat reservation, luggage fees, return discounts, promo vouchers, service fees).
- 5 Payment Methods: **Fly Express Wallet** (PIN 2580), **Mobile Money** (MTN MoMo & Airtel Money), **Cash at Stage** (15-min reservation hold), **Corporate Account**, and **Promotional Voucher**.

renderCheckout()

checkoutTotal()

renderStandardPaymentOptions()

renderPaymentPanel()

7. Digital Ticketing & Travel Pass

TICKETING

Instant pass generation, scannable security validation, and travel history.

- Post-payment confirmation screen featuring a scannable dynamic SVG/Canvas QR Code (FLYEXPRESS|TICKET|...).
- Digital Ticket Pass showing route corridor, travel date/time, vehicle plate, gate/stage, seat numbers, and driver profile.
- Lifecycle actions: Download ticket, Save to Wallet, Share ticket, Cancel & Request Refund (with cancellation retention fee calculation).
- Travel history manager with tabbed views: **Upcoming**, **Completed**, **Cancelled**.

renderSuccess()

renderTicket()

renderTickets()

renderTrips()

initTicketQr()

8. Express Parcel & Cargo Logistics

LOGISTICS

Complete 8-step stage-to-stage parcel booking, receipt generation, and status tracking.

- 8-step wizard: Sender → Recipient → Parcel Info → Delivery Service → Payment → Confirmation → Receipt → Live Tracking.
- Parcel classifications (Documents, Small/Medium/Large package, Fragile item, Business cargo) with weight, declared value, and photo upload.
- Service tiers: Standard Stage-to-Stage, Priority Express, Hold for Collection, Future Last-Mile Address Delivery.
- Digital Parcel Receipt with scannable barcode and QR code (#964201832-DL).
- Real-time tracking timeline: *Registered* → *Handed Over* → *Dispatched* → *In-Transit* → *Arrived* → *Collected*.

renderParcelBooking()

parcelPrice()

renderParcelReceipt()

renderParcelTracking()

renderParcelList()

9. Live Vehicle Tracking & Fleet Radar

FLEET OPS

Real-time vehicle position tracking, trip telemetry, and driver communication.

- OpenStreetMap interactive map with animated vehicle position marker moving along the Entebbe-Kampala corridor.
- Trip telemetry: Vehicle speed (km/h), ETA countdown timer, distance remaining, next stop announcement.
- Multi-vehicle Fleet Radar mode (tracking) showing active fleet vans across corridors.
- Driver profile modal (photo, rating, phone, plate number; 500px desktop max-width), simulated direct chat, and voice call overlay.
- One-tap Emergency SOS trigger.

renderLiveTrip()

startLiveProgress()

renderDriverProfile()

renderDriverChat()

renderDriverCall()

10. Fly Express Digital Wallet

FINTECH

In-app stored-value digital wallet and transaction ledger.

- In-memory balance tracking (e.g. UGX 32,500).
- Deposit / Top-up via Mobile Money, P2P Money Transfer, and Cash Withdrawal requests.
- Wallet PIN security (4-digit PIN verification 2580).
- Filterable transaction ledger (Deposits, Tickets, Parcels, Luggage, Refunds) with modal transaction receipts.

renderWallet()

renderTransaction()

topUpWallet()

11. Offers, Monetization & Advertising

MONETIZATION

Internal promotional campaigns and third-party sponsored commercial ad engine.

- Internal promotions: Return trip subsidies, deposit bonuses, frequent traveler rewards, referral program.
- Third-party sponsored ad cards (e.g. *Lakeview Business Centre*, *Kampala Coffee House*) with "Sponsored" badges.
- Promotional voucher redemption engine (FLY2000, PARCEL2000).

`renderOffers()``applyVoucher()``applyParcelVoucher()`

12. Notification Center

MESSAGING

Centralized operational and transactional messaging center.

- Tabbed notification categories: Travels, Parcels, Payments, System Alerts.
- Unread badges, mark-all-read action, and channel frequency preferences modal.

`renderNotifications()``renderNotification()`

13. Help, Support & Lost Property

CUSTOMER SERVICE

Customer helpdesk, FAQ repository, and lost item claims.

- Expandable FAQ accordions (Booking, Payments, Refunds, Parcel rules).
- Lost Property reporting form with item type, trip ID, date lost, description, photo attachment preview, and status tracking.

`renderSupport()``submitLostProperty()`

14. User Profile, Settings & App Info

CONFIGURATION

Profile maintenance, saved data rosters, and compliance information.

- Profile details editing, Saved Passengers roster, and Favorite Pickup Stages.
- Language selection, accessibility controls, Privacy Policy, Terms of Service, and About Fly Express info.

`renderProfile()``renderAbout()`

3. Detailed Revenue Module Architecture

Financial Deep-Dive

The **Revenue Module** defines all monetization mechanics, dynamic pricing algorithms, ancillary fee calculations, and payment collection channels built into Fly Express.

1. Distance & Corridor Passenger Fares (Core Transit Revenue)

UGX 5,000 – 30,000

Passenger tickets are calculated based on origin-destination stage distance along defined regional corridors:

- **Entebbe ↔ Kampala (Via Kajjansi / Busega):** UGX 5,000
- **Entebbe ↔ Nambole:** UGX 7,000
- **Entebbe ↔ Masaka:** UGX 20,000
- **Entebbe ↔ Lyantonde:** UGX 25,000
- **Entebbe ↔ Mbarara:** UGX 30,000

$$\text{Base Fare} = \text{Corridor Base Rate} \times \text{Total Passengers}$$

2. Subsidized Return Trip Upsell

UGX 9,000 / seat

To lock in return trip passenger volume, the platform provides a discounted return travel bundle:

- **One-Way Fare:** UGX 5,000 per passenger.
- **Return Ticket Bundle:** UGX 9,000 per passenger (saves the passenger UGX 1,000 while securing UGX 9,000 total revenue per seat instead of UGX 5,000).

3. Ancillary Luggage & Cargo Fees

+ UGX 2,000 – 7,000

Baggage monetization relies on a tiered itemized pricing structure (`appData.luggage`):

- **Small Personal Item:** UGX 0 (Free, handbag/backpack)
- **Standard Luggage (≤ 15 kg):** + UGX 2,000 per bag
- **Large Bag / Sack (≤ 25 kg):** + UGX 4,000 per bag
- **Excess / Bulky Luggage:** + UGX 6,000 per bag (requires stage dispatcher approval)
- **Fragile Item Handling:** + UGX 7,000 per item (special placement)
- **Commercial Resale Goods:** Custom rate assessed at stage desk

$$\text{Luggage Total} = \sum (\text{Item Quantity} \times \text{Tier Rate})$$

4. Seat Choice & Reservation Fees

+ UGX 500 / seat

Passengers can choose between general assignment or interactive seat selection:

- **Best Available Assignment:** Included in base ticket fare.
- **Interactive 14-Seater Map Choice:** + UGX 500 reservation surcharge per seat.

5. Special Hire (Private Vehicle Charter & 18% VAT Surcharge) UGX 50,000 – 1,000,000+

The Special Hire pricing function `calculateSpecialHirePrice()` models private vehicle charter rates:

- **Fleet Rates:** Sedan (UGX 50k-100k/day), Noah 7-seat (UGX 70k-150k/day), Minivan 10-seat (UGX 90k-180k/day), Commuter 14-seat (UGX 110k-220k/day), Highroof 18-seat (UGX 150k-300k/day), Executive Coaster 30-seat (UGX 300k-600k/day).
- **Driver Allowance:** UGX 50,000 / day standard (+ optional Tour Guide driver at + UGX 30,000 / day).
- **Fuel Allocation:** 30% of base rate or fixed UGX 150,000 / day.
- **Government VAT:** 18% Value Added Tax calculated on subtotal.

Subtotal = Base Vehicle Rate + Driver Allowance + Fuel Fee

VAT (18%) = Subtotal × 0.18

Charter Total = Subtotal + VAT

6. Express Parcel & Cargo Logistics

UGX 7,500 – 12,000

Stage-to-stage parcel logistics rates calculated in `parcelPrice()`:

- **Standard Stage-to-Stage:** UGX 7,500
- **Hold for Collection:** UGX 8,000
- **Priority Express (Earliest Departure):** UGX 10,000
- **Future Last-Mile Delivery:** UGX 12,000
- **Handling Fee:** Fixed UGX 1,500 handling charge embedded in pricing breakdown.

7. Third-Party Sponsored Advertisements

Ad Placement Revenue

Non-fare commercial advertising engine serving paid placement cards (*Lakeview Business Centre, Kampala Coffee House*) with "Sponsored" badges to commuters.

8. Fly Express Digital Wallet Float & Retention Fees

Fintech Ecosystem

In-app stored-value wallet balance reduces third-party mobile money processing fees, captures financial float interest, and applies cancellation retention fee percentages on ticket refunds.

4. Summary Checkout Revenue Formula

Final Total

When a booking reaches checkout in `renderCheckout()`, all financial components compile into the final payable amount:

Checkout Total = max(0, Base Fare + Seat Fee + Luggage Total - Return Discount - Voucher Discount)

LINE ITEM	PRICING RULE	SAMPLE CALCULATION
Base Passenger Fare	Corridor Rate × Passengers	UGX 5,000 (1-way) / UGX 9,000 (return)
Seat Choice Surcharge	UGX 500 / seat selected	+ UGX 1,000 (2 seats)
Luggage Charges	Sum of itemized baggage tiers	+ UGX 2,000 (1 Standard Bag)
Return Discount	- UGX 1,000 / seat (when active)	- UGX 1,000
Voucher Code Discount	- UGX 2,000 (code FLY2000)	- UGX 2,000
Service & Platform Fee	Included	UGX 0
Final Payable Revenue	Calculated Total	UGX 9,000